

One-Sided, Infinite, and End-Behavior Limits

Answer Key

AP Calculus AB / BC

Quick Answers

- | | |
|--------------|-----------------------------|
| 1. 1 | 6. $-\infty$ |
| 2. 3 | 7. $\frac{3}{2}$ |
| 3. $-\infty$ | 8. $-\infty$ |
| 4. 2 | 9. 2 |
| 5. 5, 7 | 10. 2, $-\infty$, ∞ |
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1. $\lim_{x \rightarrow 2^+} \frac{|x - 2|}{x - 2}$

Absolute value has to be unwrapped with a known sign, and the $+$ superscript supplies it: for x just above 2, $x - 2 > 0$, so $|x - 2| = x - 2$.

The quotient is then $\frac{x - 2}{x - 2} = 1$ for every $x > 2$, so the limit is the constant value of that expression.

1

Grading note: the left-hand limit is -1 . A student who answers “does not exist” has answered the two-sided question that was not asked.

2. $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{x^2 - 4}$

Numerator and denominator have the same degree, so divide both by x^2 :

$$\frac{3 + \frac{5}{x} - \frac{1}{x^2}}{1 - \frac{4}{x^2}}$$

Every term with an x in the denominator goes to 0, leaving the ratio of the leading coefficients.

3

3. $\lim_{x \rightarrow 5^-} \frac{1}{x - 5}$

The numerator is the nonzero constant 1 and the denominator goes to 0, so the limit is infinite; the only question is the sign.

For x just below 5 (say 4.99), $x - 5$ is a small *negative* number, so $\frac{1}{x - 5}$ is a large negative number.

$-\infty$

4. $\lim_{x \rightarrow -\infty} \frac{4x^3 - x}{2x^3 + 7}$

Same degree top and bottom, so the end behavior is the ratio of leading coefficients — and it is the same at $-\infty$ as at $+\infty$ because the odd powers appear in both numerator and denominator and their signs cancel in the quotient.

$$\frac{4x^3 - x}{2x^3 + 7} = \frac{4 - \frac{1}{x^2}}{2 + \frac{7}{x^3}} \rightarrow \frac{4}{2}$$

2

5. Piecewise f at $x = 2$.

(a) From the left, $x < 2$, so the rule in force is $x^2 + 1$, which is a polynomial and therefore continuous: substitute. $2^2 + 1 = 5$.

5

(b) From the right, $x \geq 2$, so the rule is $5x - 3$: $5(2) - 3 = 7$.

7

(c) No. A two-sided limit exists only when both one-sided limits exist *and* agree; here they are 5 and 7, so $\lim_{x \rightarrow 2} f(x)$ does not exist and f has a jump discontinuity at $x = 2$.

Manual review: part (c) is a written justification, read by a teacher; parts (a) and (b) are machine-checked.

6. $\lim_{x \rightarrow -4^+} \frac{2x}{x + 4}$

Numerator: $2x \rightarrow -8$, a nonzero negative number. Denominator: $x + 4 \rightarrow 0$, and for x just above -4 (say -3.99) it is a small *positive* number.

Negative over small positive gives a large negative quotient.

$-\infty$

Grading note: the answer flips to $+\infty$ on the left side. Whenever a sheet asks for one side only, the sign of the denominator on that side is the whole problem.

7. $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 4}}{2x + 1}$

Divide numerator and denominator by x . Inside a square root, dividing by x means dividing by x^2 — and since $x \rightarrow \infty$ is positive, $\sqrt{x^2} = x$ with no sign flip.

$$\frac{\sqrt{9x^2 + 4}}{2x + 1} = \frac{\sqrt{9 + \frac{4}{x^2}}}{2 + \frac{1}{x}} \rightarrow \frac{\sqrt{9}}{2}$$

$\frac{3}{2}$

Grading note: for $x \rightarrow -\infty$ the same work gives $-\frac{3}{2}$, because there $\sqrt{x^2} = -x$. Getting $\frac{3}{2}$ on both sides means the sign step was skipped.

8. $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right)$

Separately the two terms give $\infty - \infty$, which decides nothing. Combine them over the common denominator x^2 first:

$$\frac{1}{x} - \frac{1}{x^2} = \frac{x-1}{x^2}$$

Now read the pieces: the numerator $\rightarrow -1$, a nonzero negative number, and $x^2 \rightarrow 0$ through *positive* values (a square is positive on both sides).

$-\infty$

9. Choose k so that g is continuous at $x = 3$.

Continuity at $x = 3$ requires the two one-sided limits to exist and to equal each other (and the function value).

From the left: $\lim_{x \rightarrow 3^-} (x^2 + k) = 9 + k$.

From the right: $\lim_{x \rightarrow 3^+} (4x - 1) = 11$, which is also $g(3)$.

Set them equal: $9 + k = 11$.

$k = 2$

Check: with $k = 2$ the left rule gives $x^2 + 2 \rightarrow 11$, matching the right rule exactly.

10. **Synthesis:** $h(x) = \frac{2x^2 - 3x}{x^2 - x - 6}$.

Factor first, because factoring decides whether a zero of the denominator is a hole or an asymptote:

$$h(x) = \frac{x(2x-3)}{(x-3)(x+2)}$$

Neither $x - 3$ nor $x + 2$ divides the numerator, so both are genuine vertical asymptotes — nothing cancels.

(a) Same degree top and bottom, so the end behavior is the ratio of leading coefficients, $\frac{2}{1}$.

2

(b) As $x \rightarrow 3^-$: the numerator $\rightarrow 3(3) = 9 > 0$; the factor $x + 2 \rightarrow 5 > 0$; the factor $x - 3$ is a small *negative* number. A positive over a small negative is a large negative quotient.

$-\infty$

(c) As $x \rightarrow 3^+$ everything is the same except that $x - 3$ is now a small positive number.

$+\infty$

(d) The graph has vertical asymptotes at $x = 3$ and at $x = -2$ (the two zeros of the denominator, neither of which cancels), and a horizontal asymptote at $y = 2$, approached at both ends by part (a). Near $x = 3$ the curve plunges to $-\infty$ on the left and climbs to $+\infty$ on the right.

Manual review: part (d) is a written description, read by a teacher; parts (a)–(c) are machine-checked.